

CPT208 Human-Centric Computing

04. Design Principles and Design Alternatives

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Accessibility and Inclusiveness

Chapter 1

The Framework of Inclusion

Rethinking the definition of "Normal" in Human-Computer
Interaction

Core Pillars of Accessibility and Inclusiveness



Empathy

Understanding the diverse needs, abilities, and complex contexts of all potential users right from the design's inception.



Usability

Ensuring that digital systems are effective, efficient, and deeply satisfying to use for everyone, not merely the statistical average.



Equity

Providing fundamentally fair access to information and digital experiences by actively removing barriers in the design phase.

Defining Disability in HCI

- **The Medical Model:** Historically viewed disability strictly as a personal health condition or deficit to be cured by doctors.
- **The Social Model:** It states disability is a mismatched interaction between a person and an unaccommodating environment.
- **The Designer's Role:** By recognizing that poor design creates barriers, technologists can take full responsibility for engineering accessible interfaces.



Universal vs. Inclusive

Distinct but Related Approaches

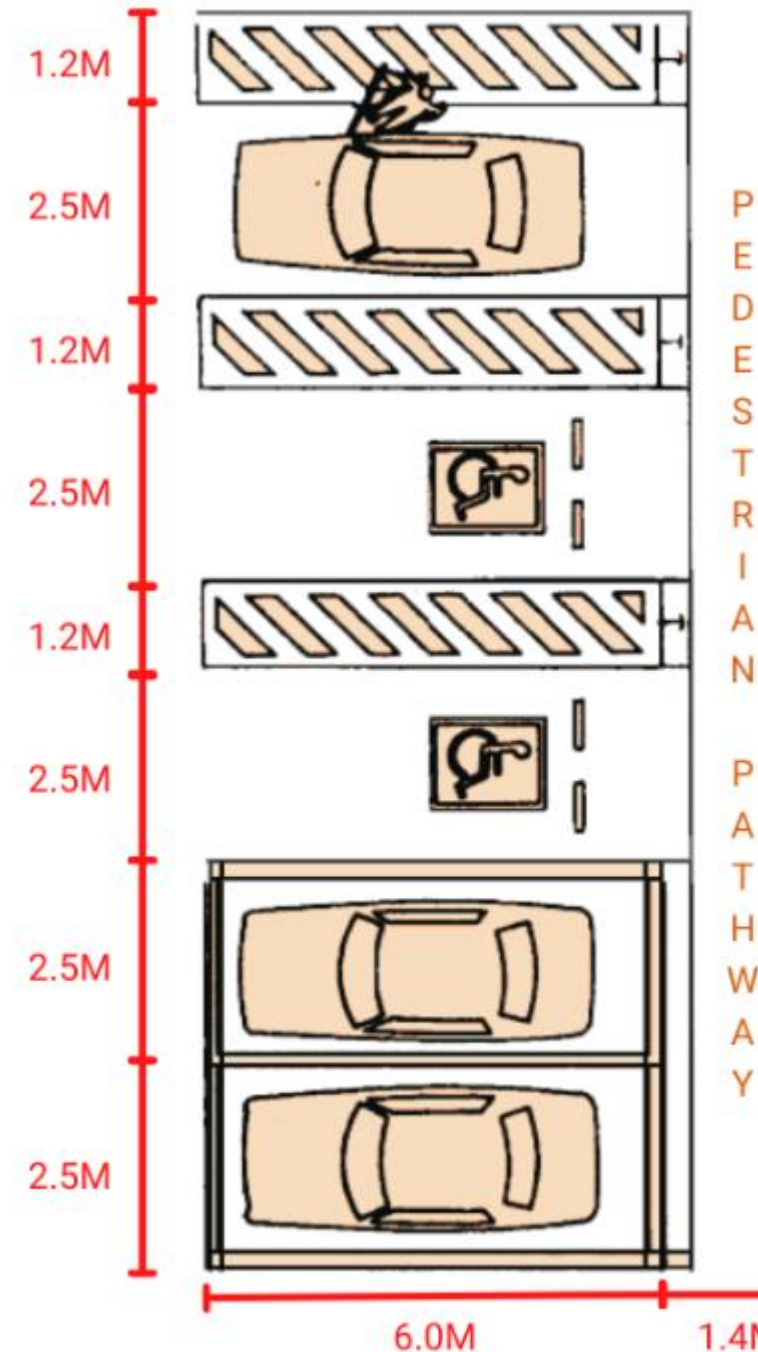
Universal Design strives to create a single optimal solution that works for everyone uniformly, much like a standard wheelchair ramp at a building entrance.

Inclusive Design recognizes that human diversity is too complex for one solution. It focuses on providing a flexible range of ways to participate.



Maximum travel
distance from
building entrance:
30M.

Loading bay width:
Maximum 3.60M.



Accessibility and inclusiveness

- **Accessibility**: the extent to which an interactive product is accessible by as many people as possible
 - Focus is on people with **disabilities**; for instance, those using android OS or apple **voiceover**
- **Inclusiveness**: making products and services that accommodate the widest possible number of people
 - For example, smartphones designed for all and made available to everyone regardless of their disability, education, age, or income

Understanding disability

- Each type can be further defined in terms of capability:
 - For example, someone might have only peripheral vision, be color blind, or have no light perception
- Impairment can be categorized:
 - **Permanent**
 - such as long-term wheelchair user
 - **Temporary**
 - that is, after an accident or illness
 - **Situational**
 - for example, a noisy environment means that a person can't hear

Designing for Assistive Tech



Screen Readers: Rely heavily on semantic HTML and labels to translate on-screen visual interfaces into coherent synthesized speech.



Braille Displays: Refreshable physical devices that absolutely require a logical document flow and precise text-based alternatives for all images.



Adaptive Switches: Essential motor accessibility tools that demand full keyboard navigation support and adequate interactive target sizing.

Principles of Inclusive Design



Recognize Exclusion Exclusion happens when we solve problems using our own personal biases. As responsible designers, we must seek out exclusion as a powerful opportunity for new ideas.



Solve for One, Extend to Many Designing an interface specifically for someone with a permanent disability consistently results in a remarkably better experience for everyone else using the platform.



Learn from Diversity Human beings are the genuine experts in adapting to diversity. Placing diverse users at the center of your design and testing process is absolutely non-negotiable.

Design Principles

Chapter 1

Design principles

- Generalizable abstractions for thinking about different aspects of design
- Derived from a mix of
 - theory-based knowledge
 - experience, and
 - common-sense

Definition & Meaning

The Definition

Design principles are fundamental, overarching heuristics and guidelines that dictate how a product should be structured, behave, and appear. They are the established rules of engagement for interaction.

The Meaning

They act as the critical bridge between human psychology and digital interfaces. By adhering to these principles, we align system mechanics with natural human cognitive models and expectations.

Primary Functions



Guide Decisions

Principles resolve team conflicts by providing an objective framework. Instead of arguing subjective preferences, teams evaluate designs against established human-centered rules.



Evaluate Quality

They serve as a measurable checklist for heuristic evaluation. Designers can assess usability flaws and identify friction points long before the product reaches user testing.



Ensure Standardization

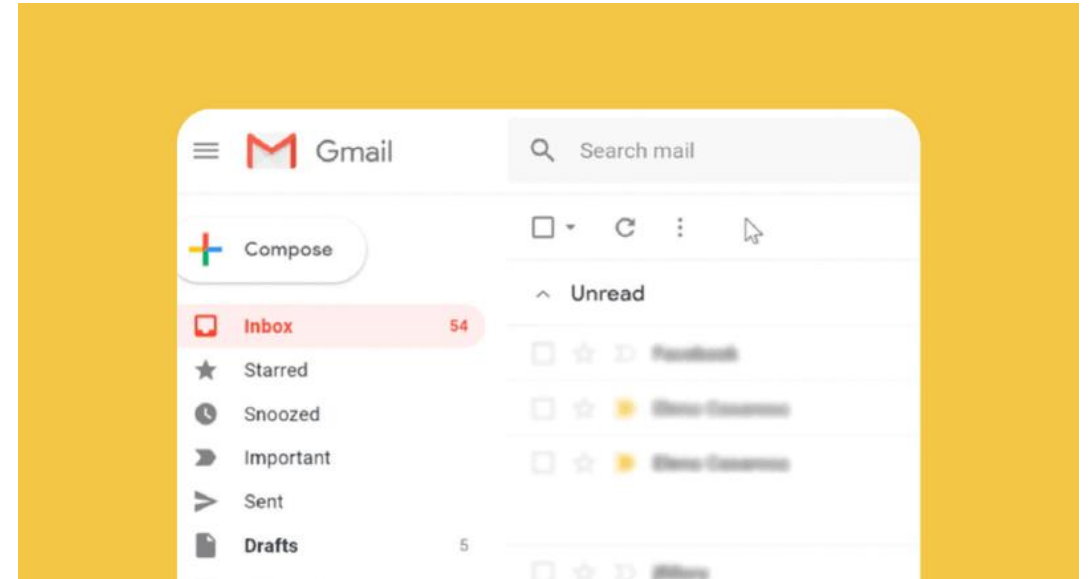
Principles enforce a uniform experience across vast ecosystems. They ensure that navigating a mobile app feels identical in logic to navigating the associated web platform.

Design principles

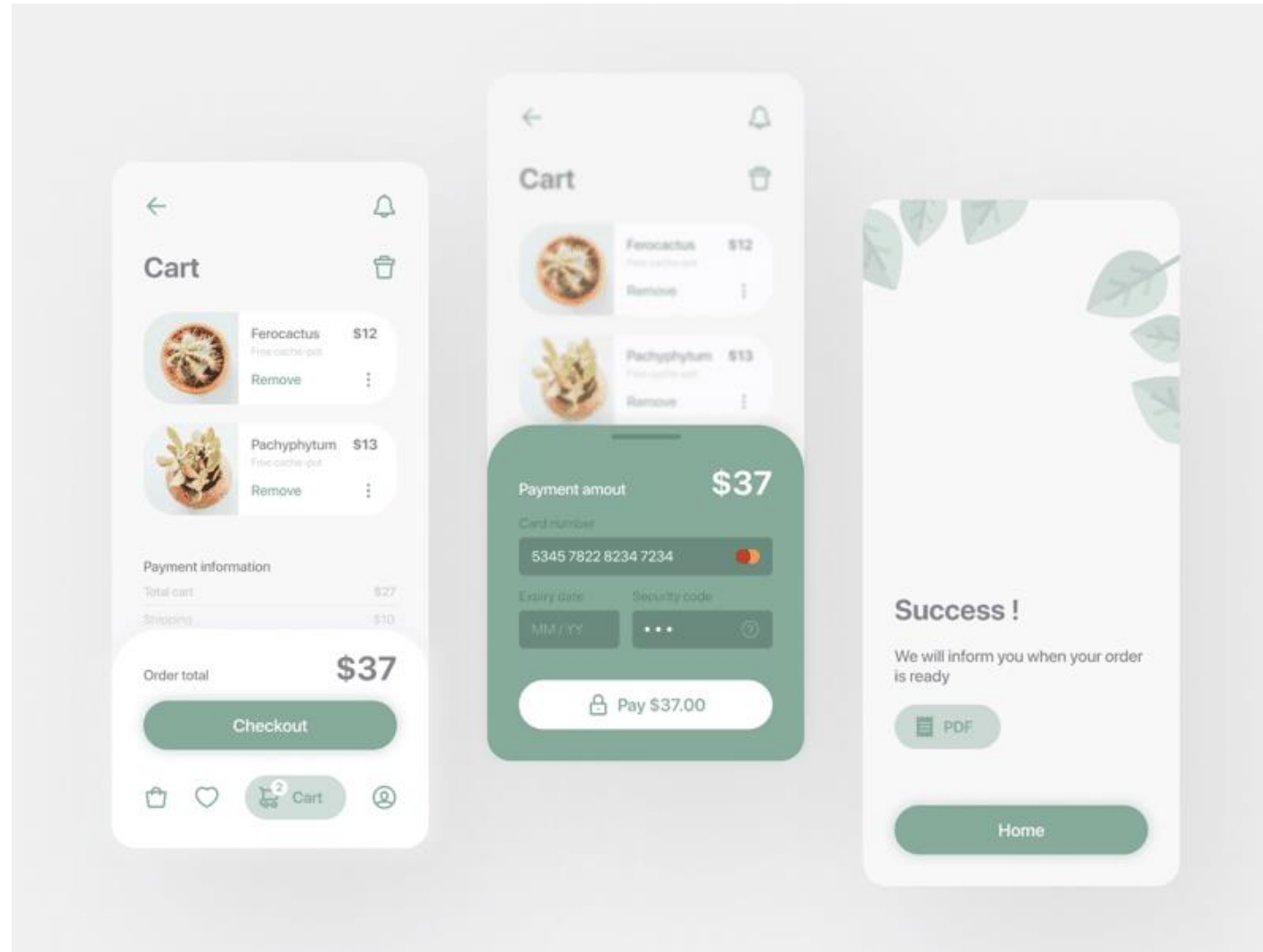
- We start with some common ones (there are more!)
 - Visibility
 - Feedback
 - Constraints
 - Consistency
 - Affordance
 - ...

Visibility

- Core Concept:
 - The more visible functions and system status are, the more likely users will be able to know what to do next. Conversely, when functions are hidden, they become significantly harder for users to discover and utilize.



Visibility

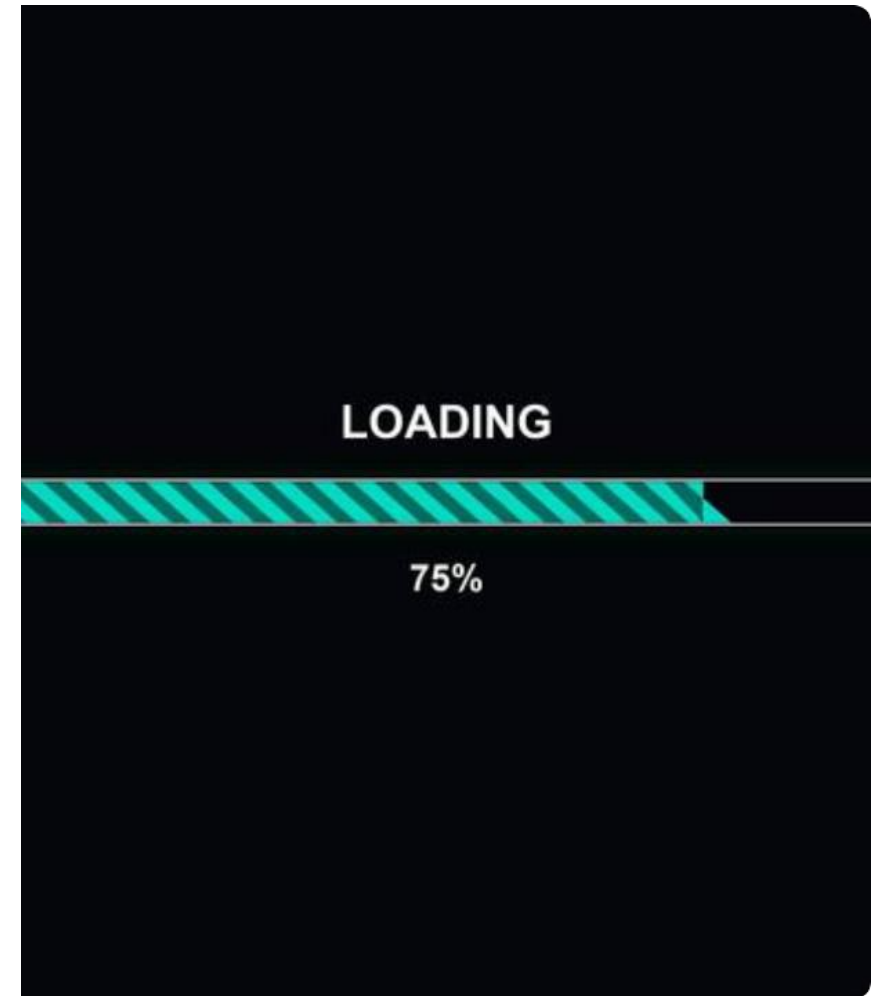


Feedback

- Keep Users Informed
- For every action a user takes, the system must provide immediate and appropriate feedback. This reassures the user that their request was received and is being processed.
- Includes sound, highlighting, animation, and combinations of these
 - For example, when screen button is clicked, it provides sound or red highlight feedback:

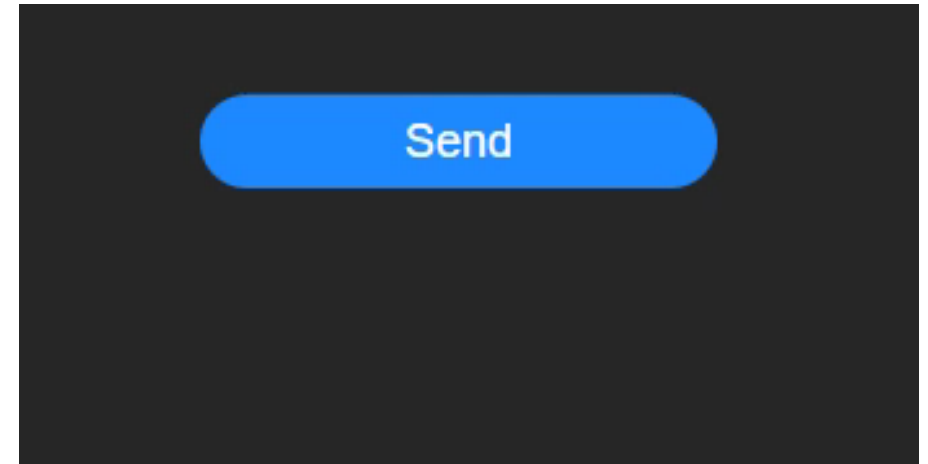
Previous → “ccclichhk”

Previous → Previous



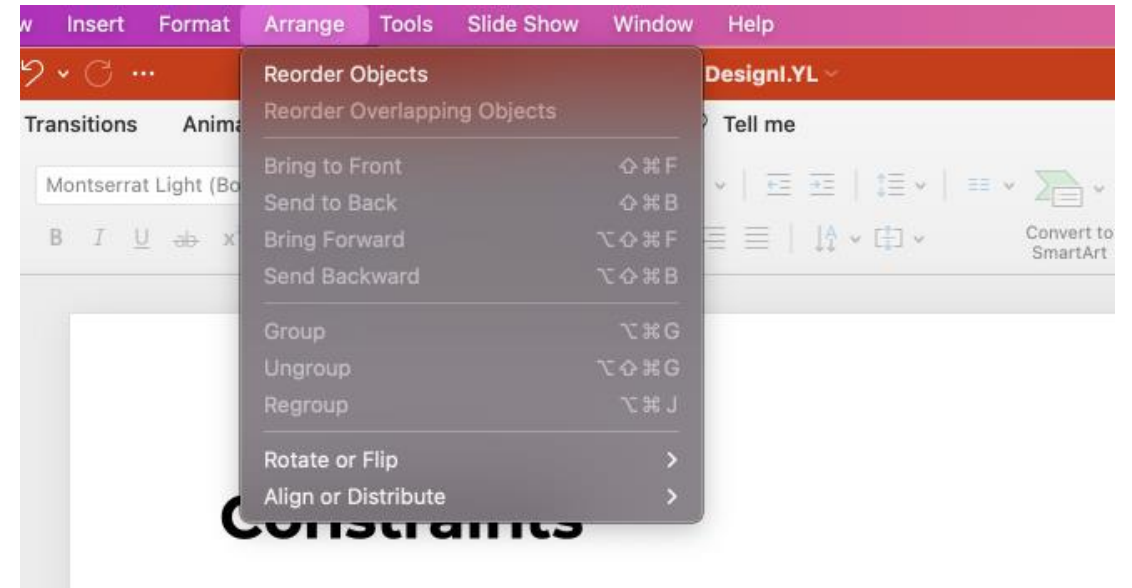
Feedback

- The "Submit" Button Lifecycle:
 - When a user clicks a "Confirm Purchase" button, it immediately changes its pressed state, then transforms into a spinning loading wheel, and finally displays a green checkmark. This continuous visual dialogue prevents the user from wondering if the click registered and stops them from clicking multiple times.



Constraints

- **Restricting** the possible actions that can be performed
- Helps prevent user from selecting incorrect options
- Physical objects can be designed to constrain things
 - for example, there being only one way you can insert a key into a lock





Constraints



Consistency

- Design interfaces to have similar operations and use similar elements for similar tasks.
 - For example, always use Ctrl key plus first initial of the command for an operation: Ctrl+c, Ctrl+s, Ctrl+o
 - **What are these and why?**
 - What about Ctrl+p, Ctrl+v, and Ctrl+x?
- The main benefit is that consistent interfaces are easier to learn and use

Consistency

- What happens if there is more than one command starting with the same letter? (for example, save, spelling, select, style)
- You have to find other initials or combinations of keys, thereby breaking the consistency rule (for example, Ctrl+s, Ctrl+Sp, Ctrl+shift+l)
- Increases learning burden on user, making them more prone to errors

Consistency



Think more about shortcuts

- Ask yourself
 - Do you use these shortcuts?
 - If so, how did you learn and memorise these?
- Ask your parents and grandparents
 - Do they use these shortcuts?
 - How easy was it for them to learn these shortcuts?
 - Again, cultural differences!

Internal consistency

- **Internal consistency** refers to designing operations to behave the same **within** an application
 - Consistency with other elements in the system
 - Cultivates a sense of orientation and trust
 - Indicate the system is well thought out and planned
 - Difficult to achieve with complex interfaces



External consistency

- **External consistency** refers to designing operations, interfaces, and so on to be the same **across** applications and devices
 - Consistency with other elements in the environment
 - Extends the benefits of internal consistency across multiple, independent systems
 - More difficult to achieve because different systems rarely observe the same design standards

Finder File Edit View Go Window Help

Safari File Edit View History Bookmarks Window Help

Photoshop File Edit Image Layer Select Filter View Window Help

External consistency

- External consistency also refers to the meaning and action are consistent to improve learnability and understanding
 - Consistent use of symbols to represent similar concepts, leverages prior knowledge and makes new things easier to learn
 - Traffic always turns yellow before red



Keypad numbers layout

→ A case of external inconsistency

(a) phones, remote controls

1	2	3
4	5	6
7	8	9
	0	

(b) calculators, computer keypads

7	8	9
4	5	6
1	2	3
0		

Affordances

- Affordance: The physical or digital properties of an object that show users what actions are possible.
- Signifiers: The specific visual cues (like shadows, borders, or colors) that communicate where the action should take place.
- Example:
 - A classic "Norman Door" fails because a flat handle implies pulling, even if the door requires pushing.
 - In UI, a button must possess a signifier (like depth or an outline) to signal that it affords clicking.



Affordances

- How to open it?
- What about this?
- It's also related to people's **mental models.**



Affordances

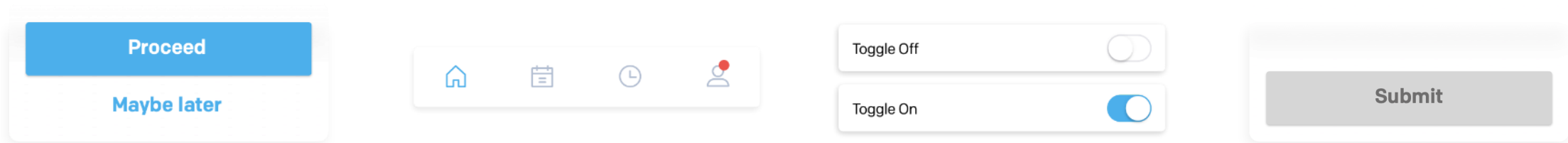
- Interfaces are virtual and do not have affordances like physical objects
- Norman argues that it does not make sense to talk about interfaces in terms of 'real' affordances
- Instead, interfaces are better conceptualized as 'perceived' affordances:
 - Learned conventions of arbitrary mappings between action and effect at the interface
 - Some mappings are better than others

Affordances



Virtual Affordances

- How do these screen objects afford?
- What if you were a novice user?
- Would you know what to do with them?



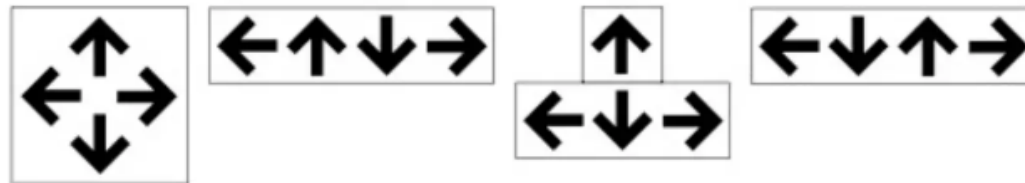
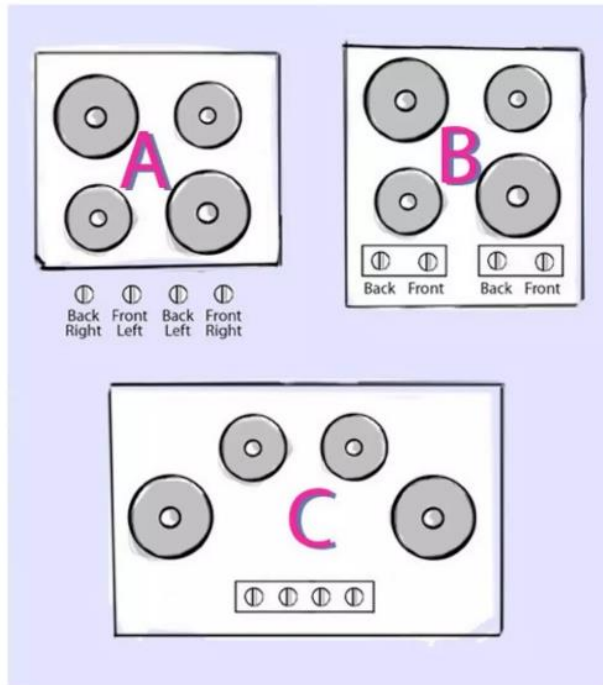
Design principles

- We talked about
 - Visibility
 - Feedback
 - Constraints
 - Consistency
 - Affordance
- There are a lot more
 - Natural mappings
 - Mental model
 - ...



Mapping

→ Relationship between controls and their effects



Shneiderman's Eight Golden Rules

Chapter 1

Shneiderman's Eight Golden Rules of Interface Design

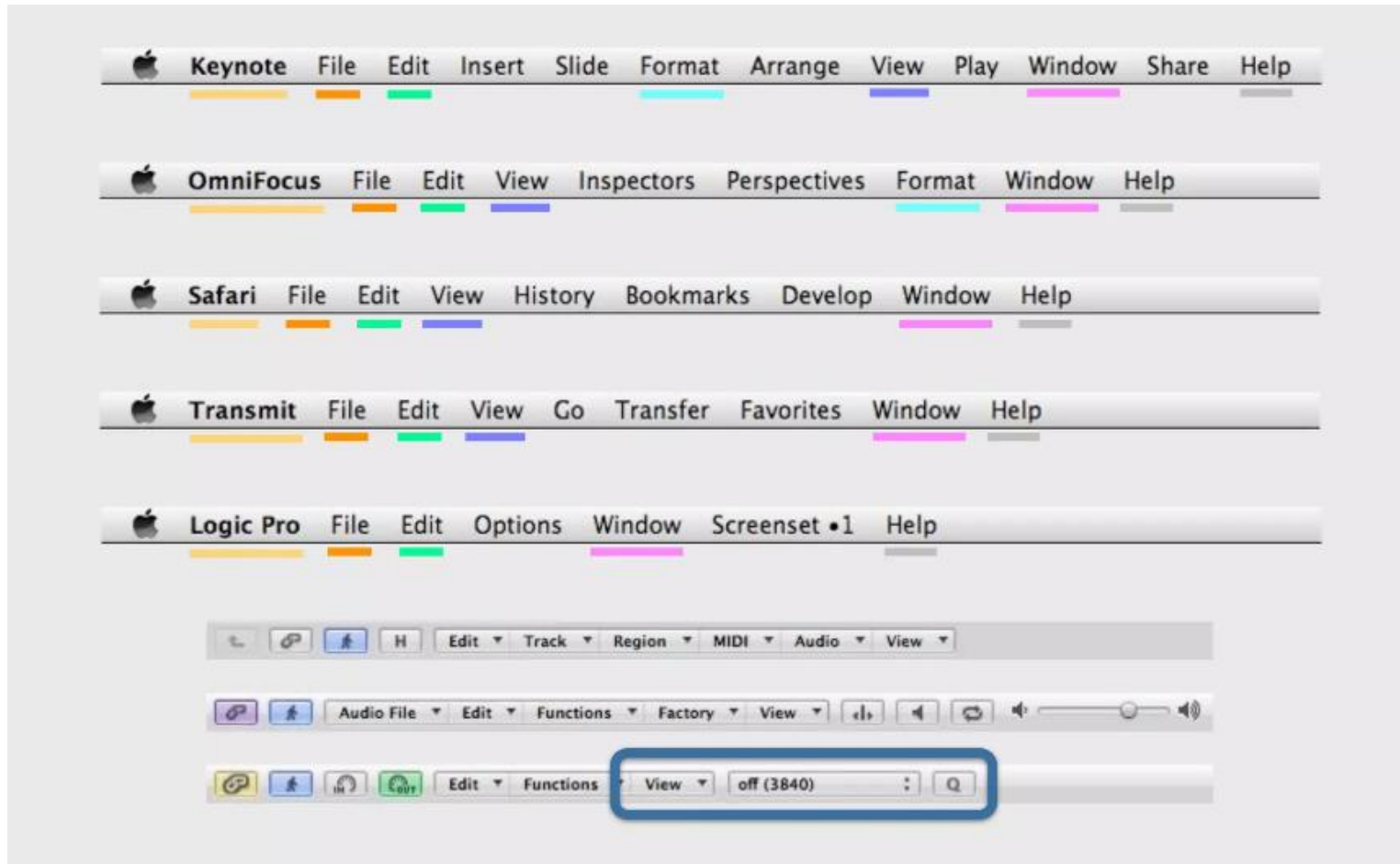
1. Strive for consistency
2. Enable frequent users to use shortcuts
3. Offer informative feedback
4. Design dialog to yield closure
5. Offer simple error handling
6. Permit easy traversal of actions
7. Support internal locus of control
8. Reduce short-term memory load

See more details in <https://www.interaction-design.org/literature/article/shneiderman-s-eight-golden-rules-will-help-you-design-better-interfaces>

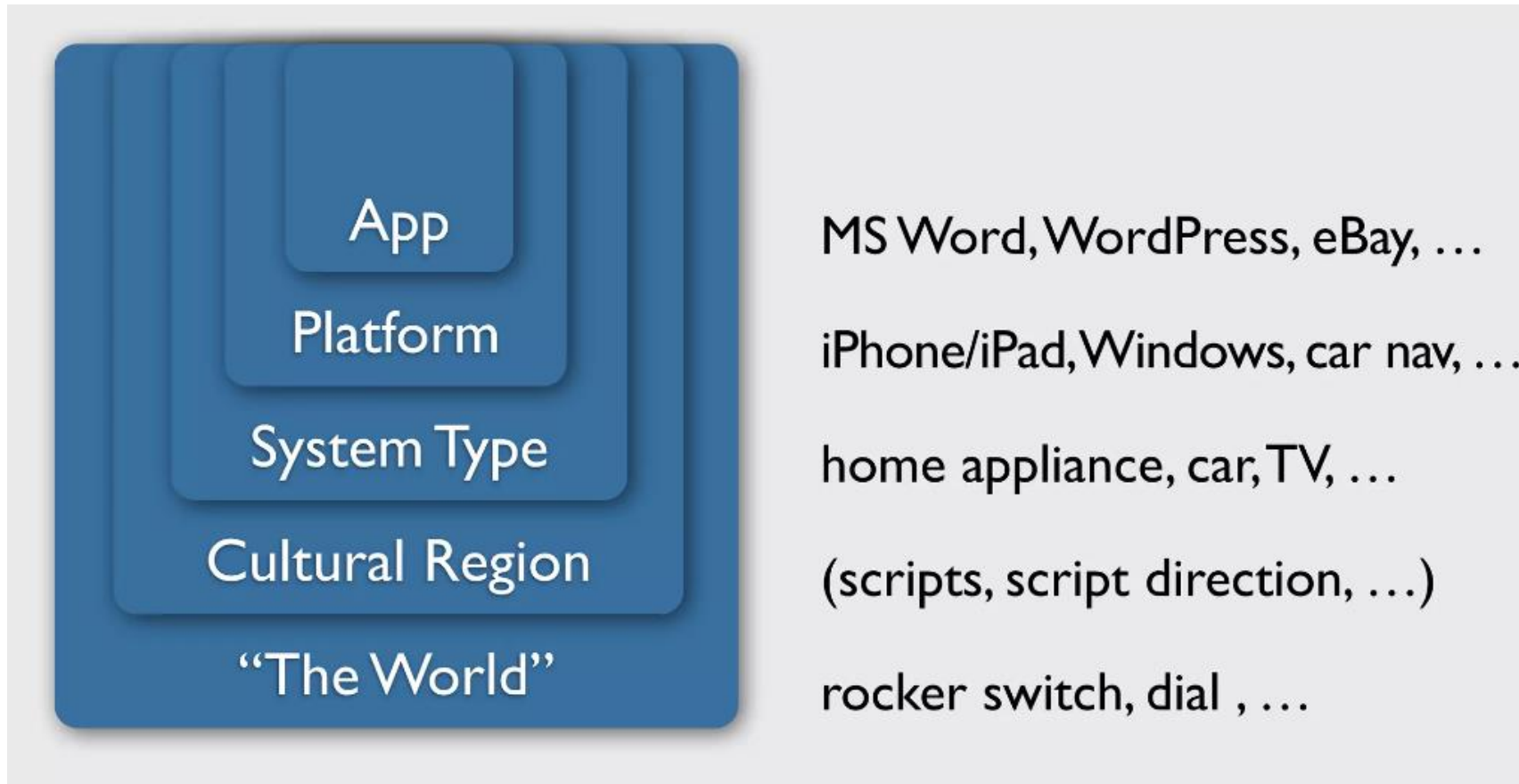
Strive for consistency

- Consistent sequences of actions should be required in similar situations; identical terminology should be used in prompts, menus, and help screens; and consistent commands should be employed throughout.
 - Workflows/Processes
 - Functionality
 - Appearance
 - Terminology

Strive for consistency



Strive for consistency

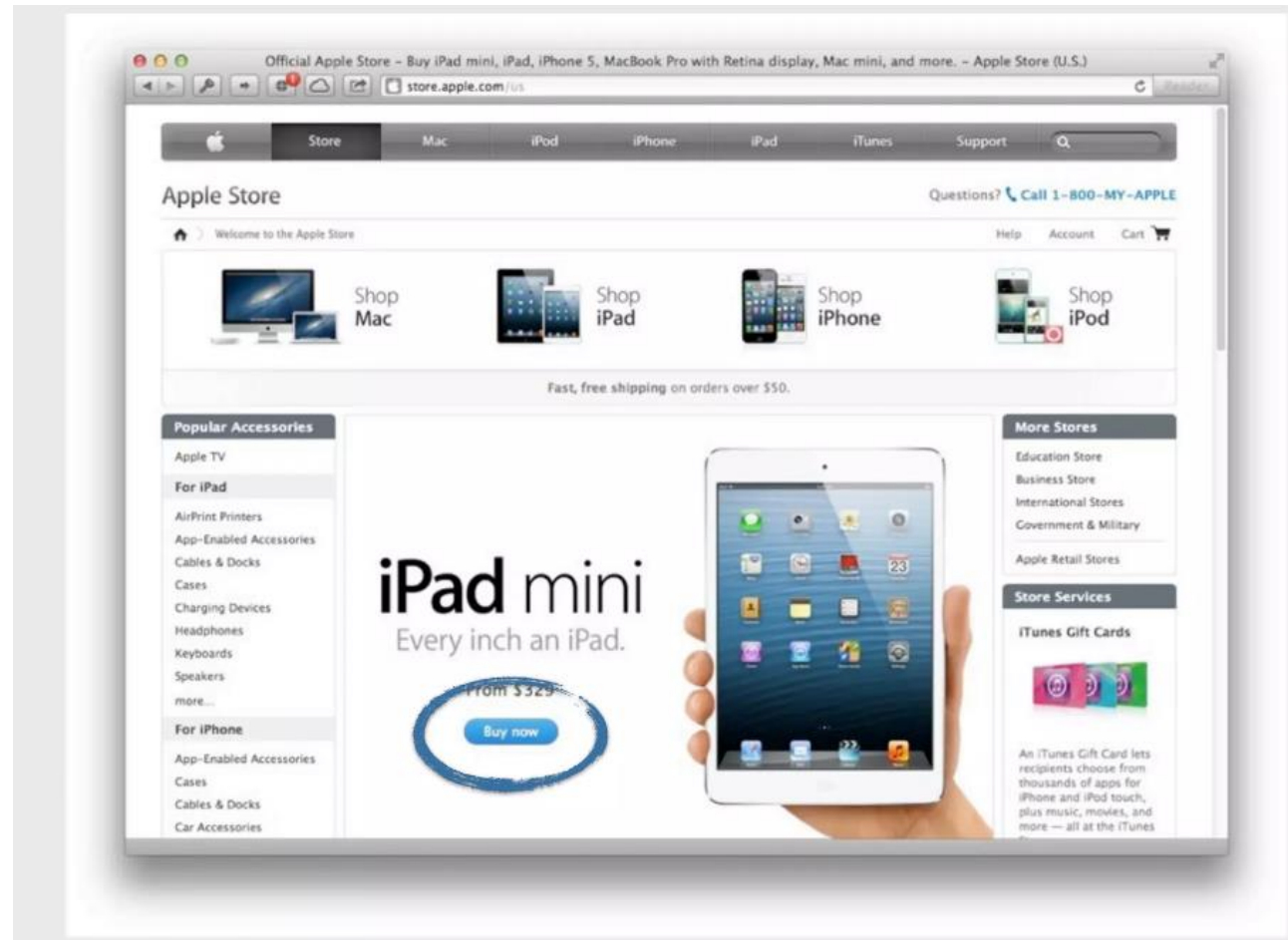


Strive for consistency



“information scent”

– consistent terminology



“information scent”

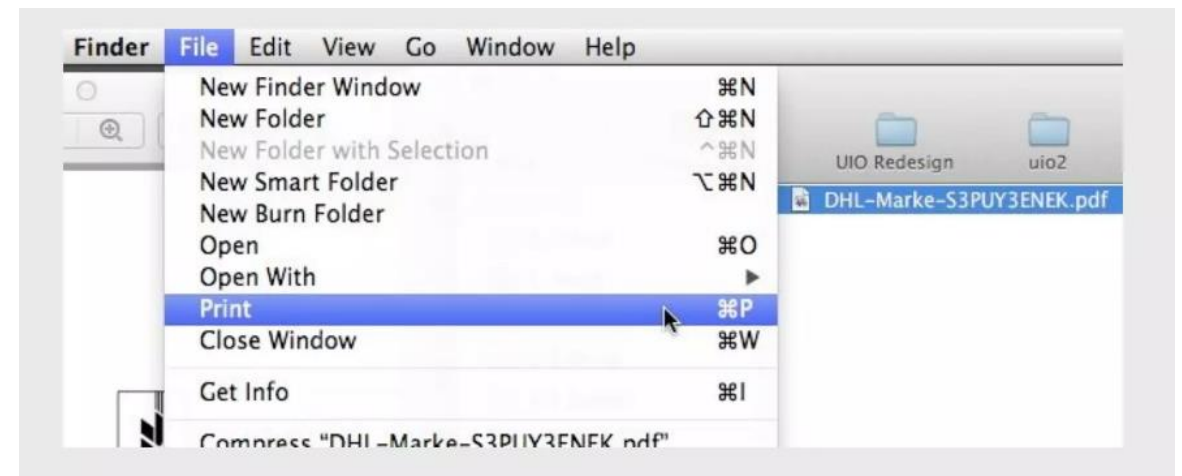
– consistent terminology



Enable frequent users to use shortcuts

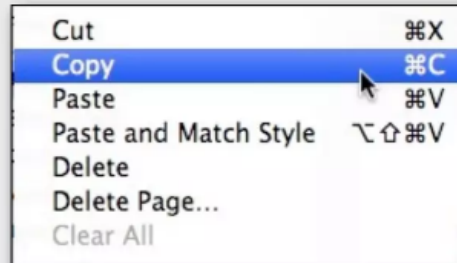
→ As the frequency of use increases, so do the user's desires to reduce the number of interactions and to increase the pace of interaction. Abbreviations, function keys, hidden commands, and macro facilities are very helpful to an expert user.

- Keyboard shortcuts
- Hidden “power user” features
- Automation



Enable frequent users to use shortcuts

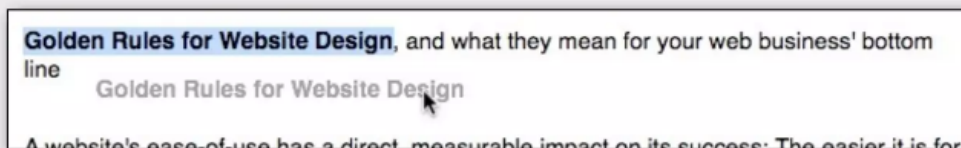
Level 1



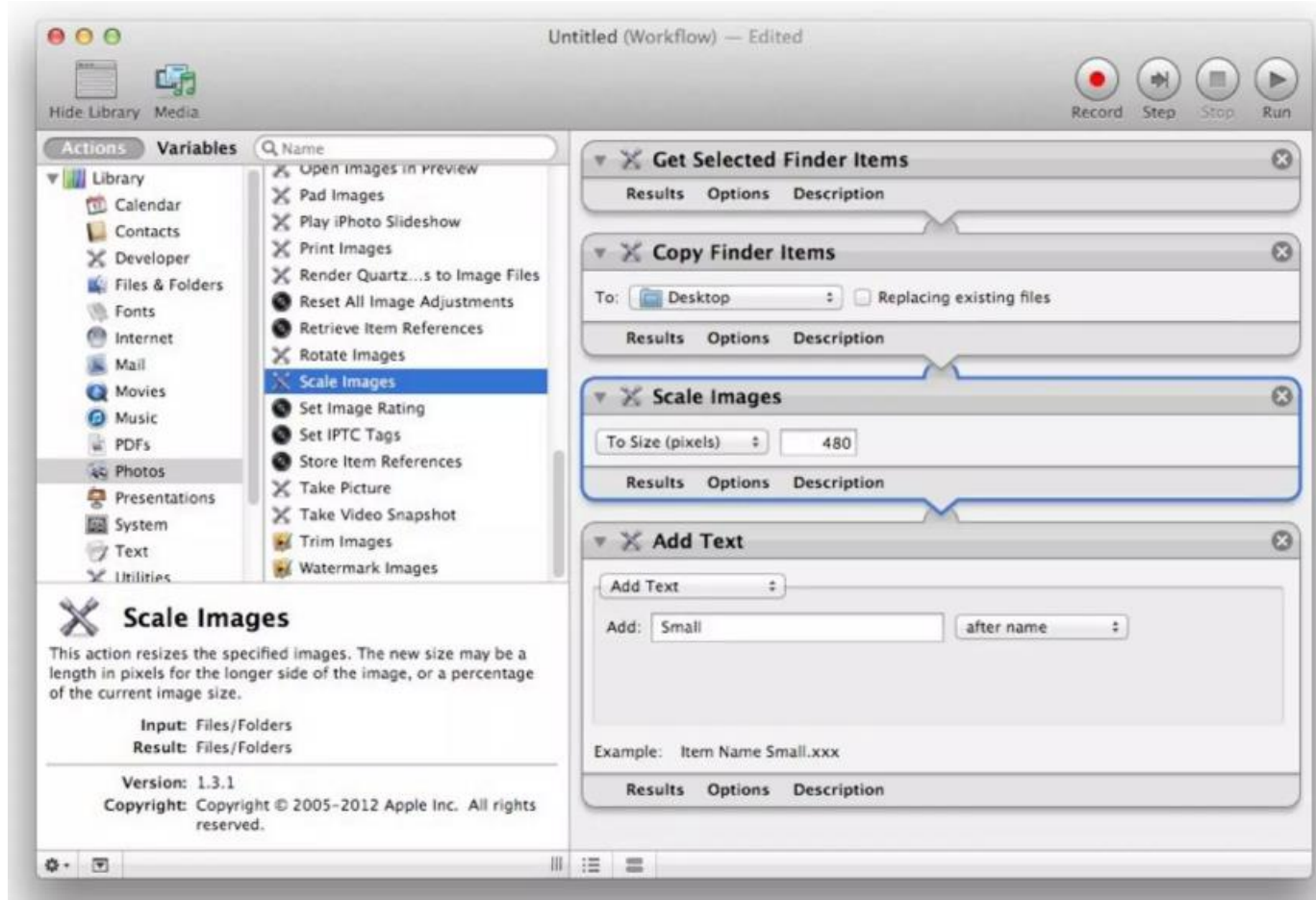
Level 2

⌘C / ⌘V

Level 3



Enable frequent users to use shortcuts



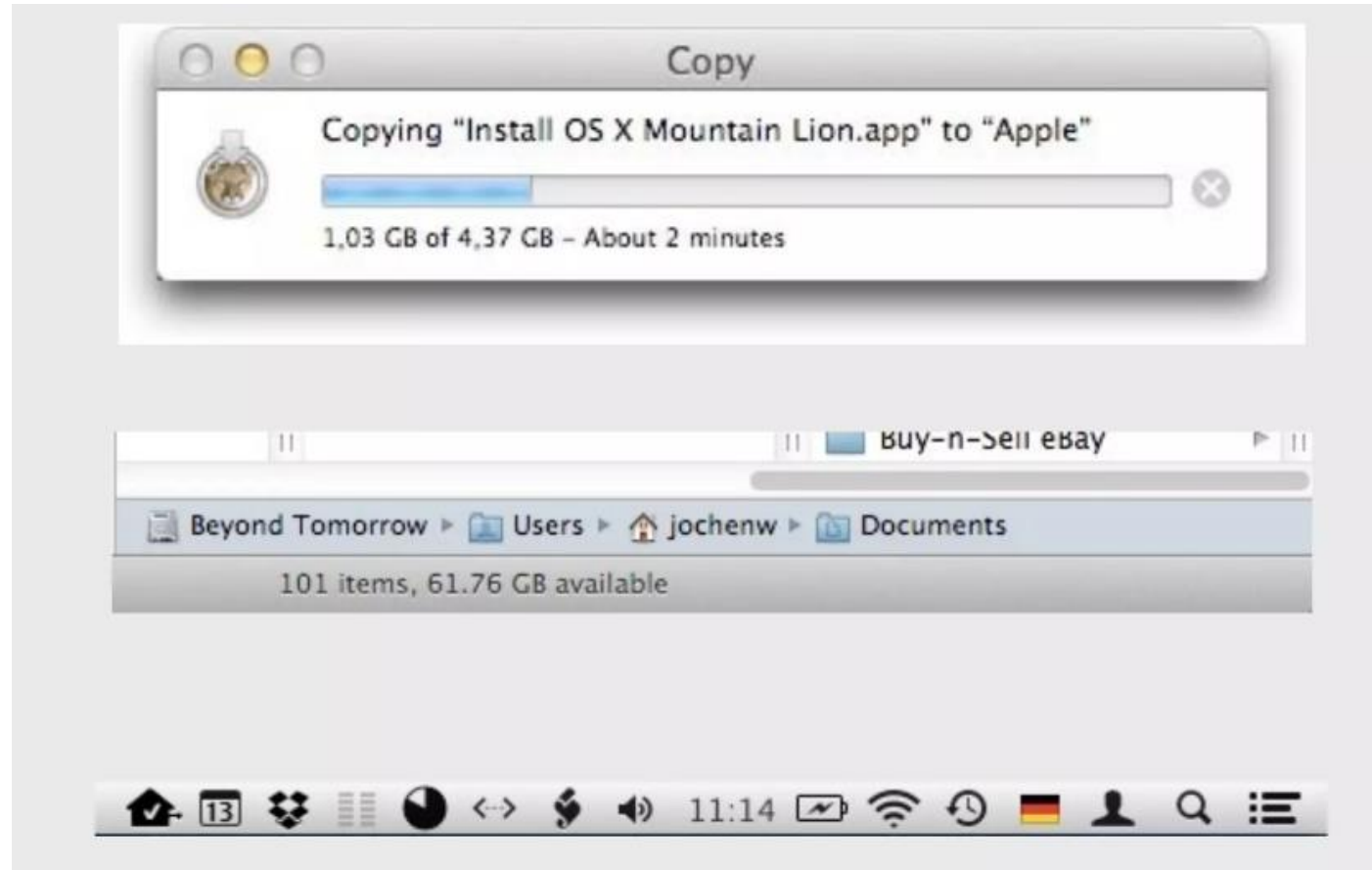
Offer informative feedback

- For every operator action, there should be some system feedback. For frequent and minor actions, the response can be modest, while for infrequent and major actions, the response should be more substantial.
 - Relevant
 - Fits importance and urgency
 - Comprehensible and meaningful
 - Within appropriate context (time and place)

Offer informative feedback

0,1 s	1,0 s	10 s
Experiencing cause and effect	Taking turns in a conversation	Typical human attention span
Respond to mouse click, key press, ...	Open window, bring up progress bar / spinner, ...	Wake machine, load file into app, start printing, ...

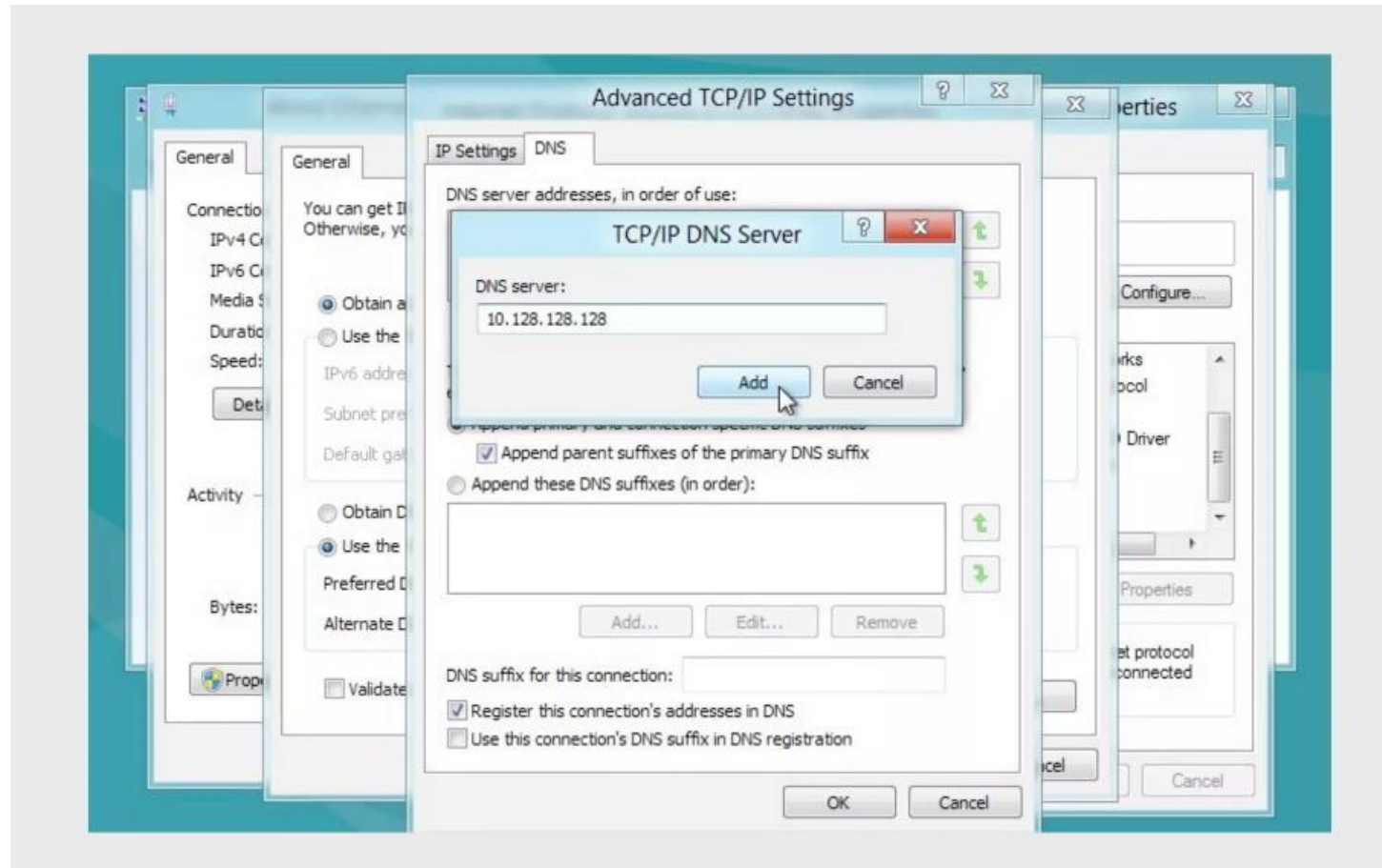
Offer informative feedback



Design dialog to yield closure

- Sequence of actions should be organized into groups with a beginning, middle, and end. The informative feedback at the completion of a groups of actions gives the operators the satisfaction of accomplishment, a sense of relief, the signal to drop contingency plans and options from their minds, and an indication that the way is clear to prepare for the next group of actions.
 - Grouping of actions
 - Explicit completion of an action
 - Well-defined options for the next step

Design dialog to yield closure



Design dialog to yield closure



cart >> checkout >> receipt

safe easy fast

Cancel and Continue Shopping

step 1: your email

Email* jochen@polytropia.com

That email address doesn't belong to any previously saved accounts. Please select "create an account" below if you'd like to save your details for your next visit.

☐ Checkout as a Guest

☒ Create an Account

Password

Retype Password

item	quantity	price
AKG Q460 Quincy Jones -Black Code: 002-012-0461 Weight: 1.1 LBS	1	\$109.00
Subtotal:		\$109.00
Shipping & Handling:		\$6.00
Order Total:		\$115.00

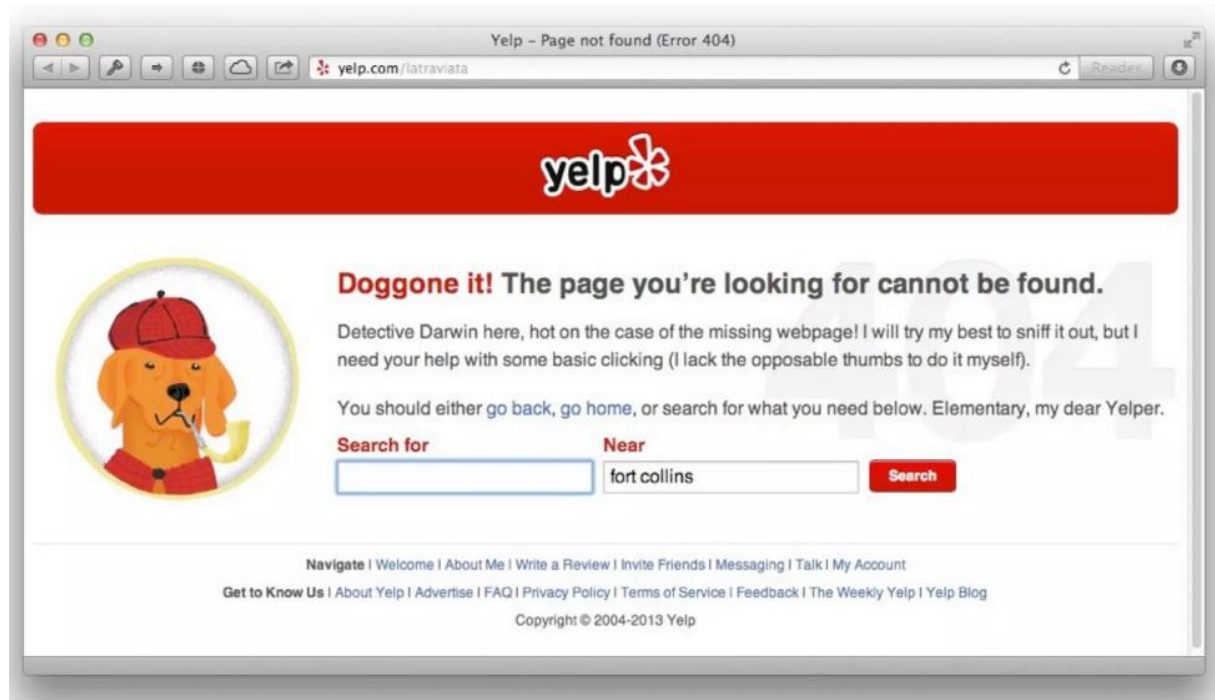
step 2: your billing address

Offer simple error handling

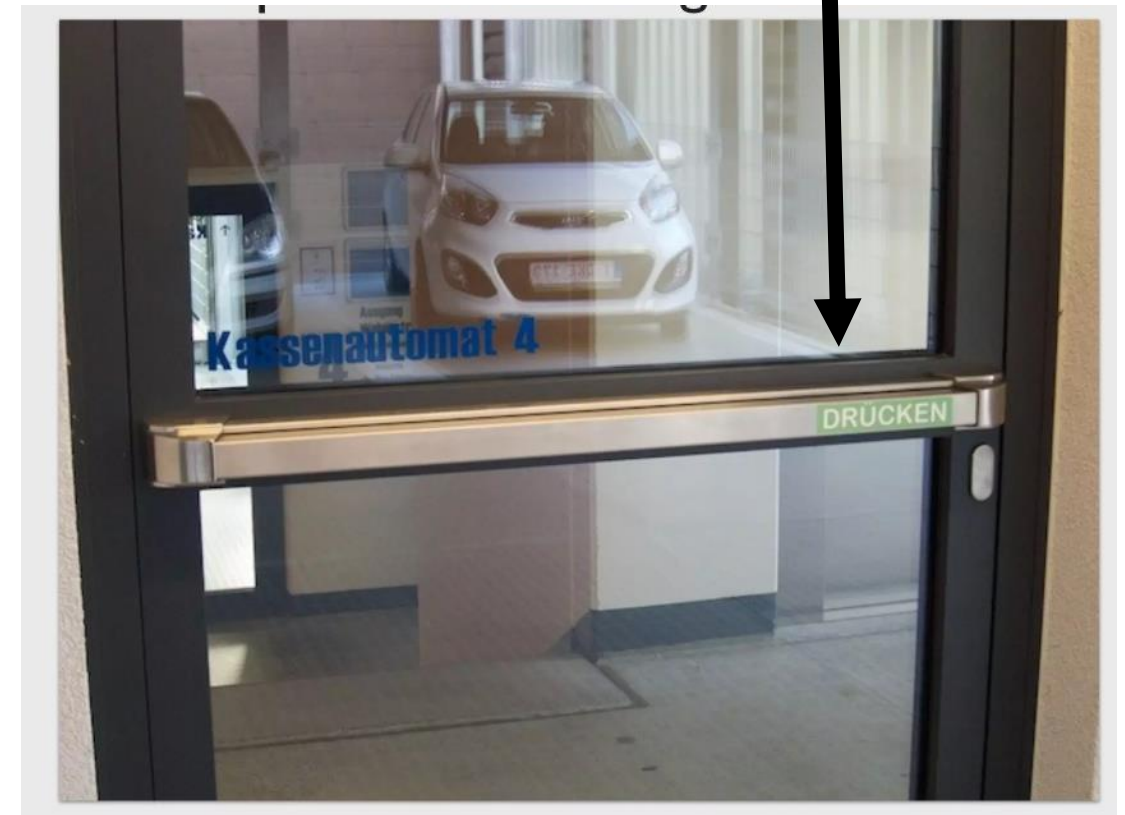
- As much as possible, design the system so the user cannot make a serious error. If an error is made, the system should be able to detect the error and offer simple, comprehensible mechanism for handling the error
 - Error prevention over error correction
 - Automatic detection of errors
 - Clear error notifications
 - Hints for solving the problem



Offer simple error handling



Press



Offer simple error handling

The image shows a user interface for creating an Apple ID. It features two main sections: 'Your Name' and 'Apple ID and Primary Email Address'. The 'Your Name' section contains the text 'Andy Geschäftsleitung' and an 'Edit' link. The 'Apple ID and Primary Email Address' section contains a text input field with the value 'abc@something.com'. Below the input field, there is a paragraph of text explaining the use of the Apple ID. A validation error message is displayed in a white box with a grey border, pointing to the input field. The message states 'Apple ID must:' followed by three bullet points: 'Be a valid email address', 'Not already be in use', and 'Not use a domain owned by Apple'.

Your Name
Andy Geschäftsleitung [Edit](#)

Apple ID and Primary Email Address

You'll use your new Apple ID to sign in to services such as iTunes, the App Store, the iBooks Store, the Mac App Store, the Online Store, and Game Center. Please note that you might be required to verify your email address before you can start using your new Apple ID.

Apple ID must:

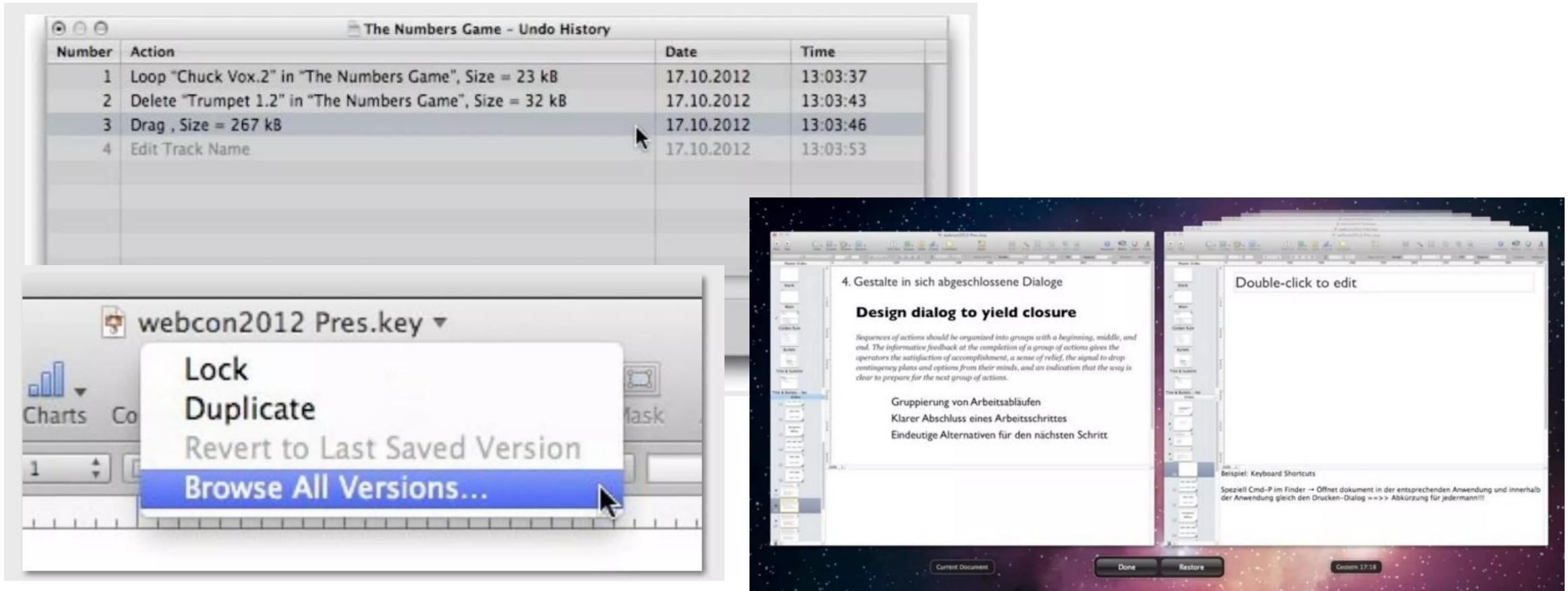
- Be a valid email address
- Not already be in use
- Not use a domain owned by Apple

Permit easy reversal of actions

- This feature relieves anxiety, since the user knows that errors can be undone; it thus encourages exploration of unfamiliar options. The units of reversibility may be a single action, a data entry, or a complete group of actions.
 - No interference with workflow
 - More freedom for the user
 - Single-action undo vs. action history



Permit easy reversal of actions

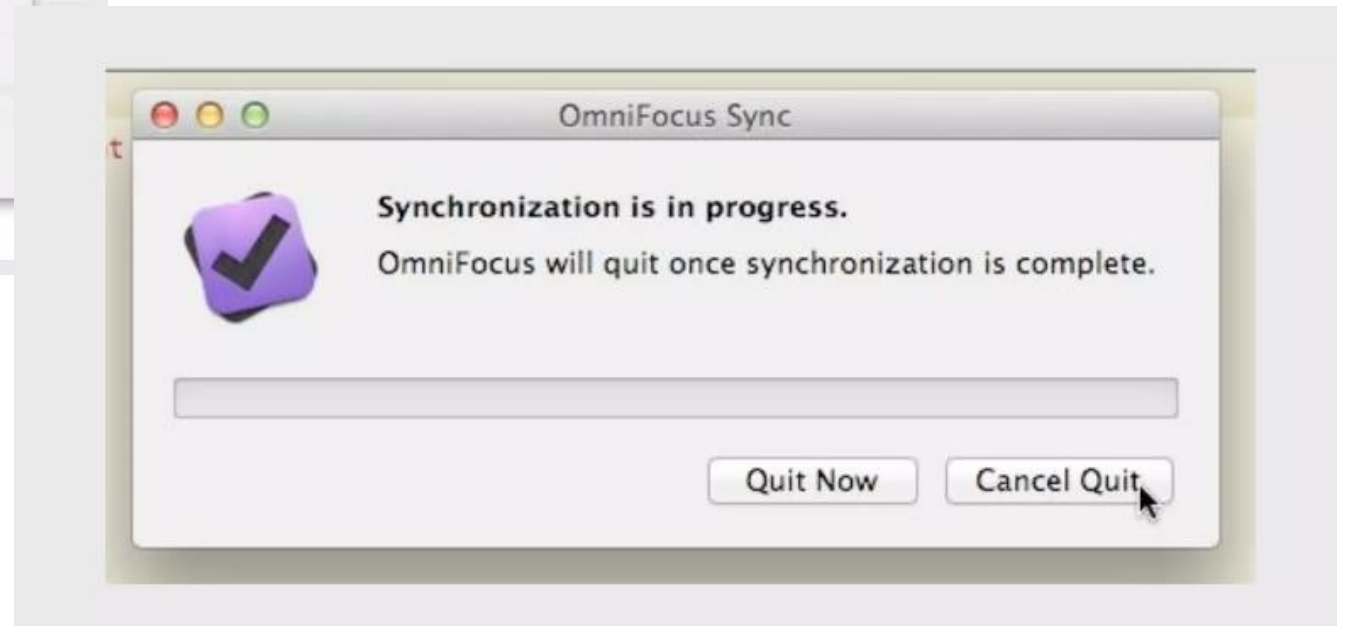


Support internal locus of control

- Experienced operators strongly desire the sense that they are in charge of the system and that the system responds to their actions. Design the system to make users the initiators of actions rather than the responders
 - The user commands, the system obeys
 - Strongly relies on the informative feedback
 - “The principle of least surprise”



Support internal locus of control



Support internal locus of control

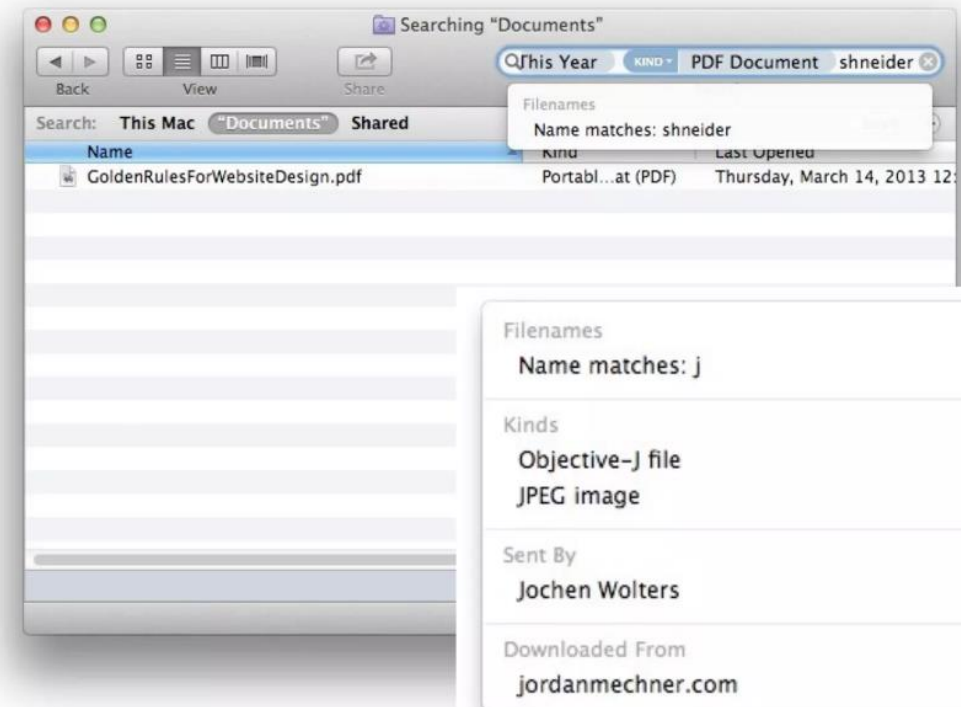
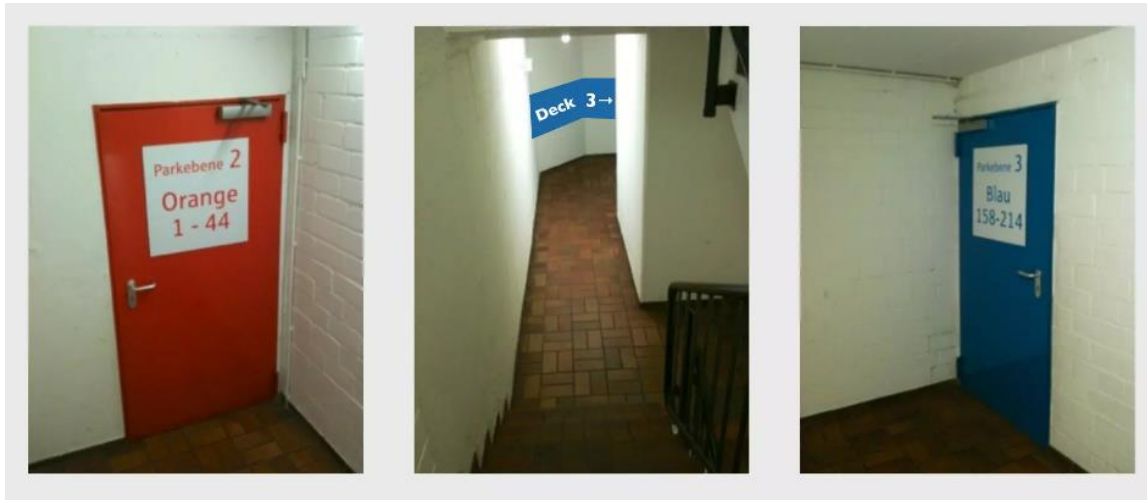


Reduce short-term memory load

- The limitation of human information processing in short-term memory requires that displays be kept simple, multiple page displays be consolidated, window-motion frequency be reduced, and sufficient training time be allotted for codes and sequences of actions.
 - Clear structure: windows, dialogs, app in its entirety
 - “Recognition over recall”
 - Implicit help

Understanding a simple sentence can — if interrupted with a tangent like this one, which contains just twenty words, but already noticeably challenges your short-term memory — become a problem.

Reduce short-term memory load



Alternative designs

Chapter 2

The Dual Process of Design

1. Divergent Thinking

This is the phase of expansion. The goal is to generate as many wildly different ideas and alternative concepts as possible without immediate judgment or technical constraint. It widens the "design space" to uncover novel approaches.

2. Convergent Thinking

This is the phase of refinement. Once a broad landscape of alternatives exists, designers evaluate, combine, and eliminate options based on design principles and user needs to converge on the single most optimal solution.

Alternative designs

- How to generate alternative designs?
- How to choose among alternatives?

How to generate alternatives?

- Humans tend to stick with something that works
- Considering alternatives helps identify better designs
- Where do alternative designs come from?
 - 'Flair and creativity': research and synthesis
 - Cross-fertilization of ideas from different perspectives
 - Users can generate different designs
 - Product evolution based on changing use
 - Seek inspiration: similar products and domain, or different products and domain
- Balancing constraints and trade-offs

Methods for Generating Alternatives



Brainstorming

Unstructured, rapid ideation sessions. The key rule here is that all criticism is deferred. Quantity is encouraged over immediate feasibility to spark creative leaps.



Crazy Eights

A core design sprint method where participants are challenged to sketch 8 distinct alternative interfaces in exactly 8 minutes. It forces designers past their initial, obvious ideas.



Morphological Analysis

Breaking a product down into its essential attributes, and then combining those attributes in novel, randomized ways to generate unexpected alternative concepts.

Evaluating the Alternatives



Heuristic Evaluation Before showing users, designers analyze the alternative concepts against established rules (like Visibility, Feedback, and Consistency) to eliminate objectively poor designs.



A/B Testing Deploying two distinct alternative designs simultaneously to different user segments. By measuring quantitative data (e.g., click-through rates), the optimal design is chosen by user behavior, not designer preference.



Qualitative User Feedback Showing prototypes to actual users. Asking them "why" they prefer Alternative A over Alternative B provides deep insights that raw data cannot capture.

Case: hotel search in different apps



→ Different ways in entering the map mode

Any Questions?

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